18/12/24 M/S

USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [22MEDE512]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

RENEWABLE SOURCES OF ENERGY

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	How does IoE improve energy efficiency?	L2	1	**02**
	b.	Define solar azimuth angle and zenith angle.	L1	2	**02**
	c.	What is meant by teetering in wind turbines?	L2	3	**02**
	d.	What are the main challenges in scaling up biomass gasifiers?	L2	4	**02**
	e.	List the merits of Ocean Thermal Energy Conversion (OTEC).	L2	5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Discuss the advantages and disadvantages of the renewable energy sources	L2	1	**05**
		OR			
	b.	Examine the environmental and economic challenges associated with conventional energy sources.	L4	1	**05**
3.	**a.**	Explain the working and applications of solar dryers.	L2	2	**05**
		OR			
	b.	Write short Discuss the working principle of photovoltaic (PV) cells with sketch.	L2	2	**05**
4.	**a.**	Explain with neat sketch the working of a wind energy system with main components.	L2	3	**05**
		OR			
	b.	Compare and contrast between horizontal axis and vertical axis wind mills.	L4	3	**05**
5.	**a.**	Describe in detail on the following processes. **i.** Pyrolysis **ii.** Carbonization	L2	4	**05**
		OR			
	b.	Compare and contrast downdraft, updraft, and fluidized bed gasifiers with respect to operating principles, efficiency, and suitability for different biomass feed stocks.	L4	4	**05**
6.	**a.**	List the sources of biomass energy and analyze the methods used for harvesting.	L4	4	**05**
		OR			
	b.	Discuss the energy scenario in India.	L2	5	**05**
7.	**a.**	List and brief on the criteria to be considered for site selection in OTEC power plants.	L4	5	**05**

Learning Resource Centre Jain Global Campus

		OR			
	b.	Write short notes on small, mini, and micro hydroelectric systems.	L4	5	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Analyze the role of renewable energy in achieving economy and sustainable development goals (SDGs).	L4	1	**10**
		OR			
	b.	Evaluate the impact of renewable energy adoption on energy security and its implications for national and global energy policies.	L4	1	**10**
9.	**a.**	Analyze the principles and benefits of solar distillation and solar stills. How do these technologies provide solutions for water scarcity in remote areas?	L4	2	**10**
		OR			
	b.	Explain the working of Graphene Solar PV technology with neat sketch stating the merits and demerits.	L2	2	**10**
10.	**a.**	Describe the working of horizontal axis wind turbine, stating their components. Also discuss it's pros and cons.	L2	3	**10**
		OR			
	b.	What are the recent developments and technological advancements in wind turbine design and wind energy systems and how do they contribute to improving efficiency and reducing costs?	L4	3	**10**
11.	**a.**	Describe the process of biogas and biochar production using pyrolysis, and discuss their applications in energy and agriculture.	L3	4	**10**
		OR			
	b.	Discuss the working of KVIC biomass gasifier with suitable sketch stating the merits and demerits.	L3	4	**10**
12.	**a.**	Write short notes on single basin and double basin tidal energy conversion systems.	L2	5	**10**
		OR			
	b.	Describe the working of hydrogen fuel cell with suitable sketch stating their merits and demerits.	L2	5	**10**
13.	**a.**	Describe the different types of geothermal resources and their suitability for power generation.	L2	5	**10**
		OR			
	b.	Discuss the various solar passive heating and cooling techniques used in building design. How do these techniques contribute to energy efficiency?	L2	2	**10**